STANDATA variance 23-BCV-007

Building

Maximum span values for joists, rafters and beams

Date Issued: May 2024

Purpose

To recognize the acceptability of the values within the “The Span Book” published by the Canadian Wood Council (CWC).

Discussion

Article 9.23.4.2. of the National Building Code - 2023 Alberta Edition (NBC(AE)) references the span tables located at the end of Part 9 for determining the maximum permissible spans of joists, rafters, and beams under various loading conditions. These tables are formulated by the CWC on behalf of the National Research Council of Canada, as noted in the Appendix Note A-9.23.4.2.

Code References

Article 9.23.4.2.

9.23.4.2. Spans for Joists, Rafters and Beams
(See Note A-9.23.4.2.)
1) Except as required in Sentence (2) and Article 9.23.14.10., spans for wood joists and rafters shall conform to the spans shown in Span Tables 9.23.4.2.-A to 9.23.4.2.-G for the uniform *live loads* shown in the Tables. (See Article 9.4.2.2.)
2) Spans for floor joists that are not selected from Span Tables 9.23.4.2.-A and 9.23.4.2.-B and that are required to be designed for the same loading conditions, shall not exceed the design requirements for uniform loading and vibration criteria.
(See Note A-9.23.4.2.(2).)
3) Spans for built-up wood and glued-laminated timber floor beams shall conform to the spans in Span Tables 9.23.4.2.-H to 9.23.4.2.-K. (See Article 9.4.2.2.)

Appendix Note A-9.23.4.2.

A-9.23.4.2. Span Tables for Wood Joists, Rafters and Beams.

The Canadian Wood Council publishes “The Span Book,” a compilation of span tables covering many of these alternative configurations. Although these tables have not been subject to the formal committee review process, the Canadian Wood Council generates, for the CCBFC, all of the Code's span tables for wood structural components; thus Code users can be confident that the alternative span tables in “The Span Book” are consistent with the span tables in the Code and with relevant Code requirements.

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Application

This variance applies to the alternative configuration of joists, rafters, and beams available in 'The Span Book' published by CWC.

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

The values for the maximum spans of joists, rafters, and beams referenced in Article 9.23.4.2. of the NBC(AE) are permitted to be determined using the values from:

The Span Book (2020 Edition)
ISBN: 978-1-989039-06-9
available for purchase, published and distributed by:

The Canadian Wood Council
99 Bank Street, Suite 420
Ottawa, Ontario K1P 6B9

Phone Number: (613) 747-5544
Fax Number: (613) 747-6264
Web: www.cwc.ca

This VARIANCE is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

Issued by the Provincial Building Administrator

[Original signed]

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